

AISHWARYA SATWANI

Doctoral Student in Computer Science $\diamond$ Value-Aligned Recommender Systems

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Primary Languages: Python, Java, JavaScript, C++ $\diamond$ Frameworks: PyTorch, TensorFlow, Hugging Face

EDUCATION

Ph.D. in Computer Science, University of Colorado Boulder

Expected 12/2027

Advisor: Dr. Robin Burke

Research Focus: Value-Based Recommender Systems, ranking, and multistakeholder optimization.

M.S. in Computer Science, University of Colorado Boulder

2021 – 2023

B.E. in Computer Science and Engineering, B.M.S. College of Engineering

2014 – 2018

RESEARCH EXPERIENCE

Research Assistant

2021 – Present

University of Colorado Boulder

Platform for Open Recommendation and Online eXperimentation (POPROX NSF Grant #2232551)

Challenge: No infrastructure existed for longitudinal, real-world recommender systems experiments with actual news readers under naturalistic conditions.

- Co-built POPROX, an NSF-funded online experimentation platform for news recommendation research, in collaboration with 5 universities and the Associated Press, delivering daily newsletters to real subscribers and supporting controlled A/B experiments with external recommendation models.
- Built the offline evaluation framework used for iterative model development, covering accuracy metrics (NDCG, MRR), non-accuracy metrics (RBO, RBE, ILS, LIP), section based evaluation (like Section-CTR, Section RBP etc.) enabling reproducible pre-deployment evaluation across experiment teams.
- Co-Designed the candidate retrieval pipeline, ingesting 100–150 articles daily via AP APIs with topic annotations, the user preference model capturing 14 topic categories on a 5-point scale with dynamic update support and a section-based newsletter.

Personalized Diversity Experiment

Challenge: Diversity and engagement are typically at odds, increasing variety in recommendations lowers clicks, creating a "diversity tax" that discourages deployment of diverse systems.

- Designed a 5-week controlled experiment on POPROX comparing three conditions: a relevance-based baseline, generalized diversification (MMR), and personalized diversification (MMR-P) calibrated to each user's topic breadth.
- Found that generalized diversity (MMR) caused a sharp decline in clicks and participation, while personalized diversity (MMR-P) maintained engagement at baseline levels while achieving the highest measured content diversity, demonstrating that diversity and engagement are compatible when diversification is user-specific.
- Used explicit topic preferences as a proxy for "diversity tolerance" to personalize the degree of diversification per user, resolving the engagement-diversity tradeoff without requiring implicit feedback. Paper under review.

Operationalization of Multistakeholder Values

Challenge: Journalists' values and perspectives are absent from news recommender system design, no methodology existed to surface them as concrete algorithmic inputs.

- Conducted semi-structured interviews with 12 digital journalists (national, local, sports, photo, and editorial roles) to identify how current news recommender systems conflict with journalistic practice.
 - Found that journalists systematically adapt their writing to game opaque algorithms, often at the expense of journalistic integrity, and that engagement-optimized systems consistently surface sensational content over quality reporting.

- Identified opacity, lack of editorial agency, and the privileging of virality as core failure modes of current systems, establishing the value dimensions (timeliness, accountability, transparency, public service) that a journalist-aligned RS should encode.
- Currently conducting 4 virtual focus groups with 12 journalism students and professionals.
 - This study analyzes how editorial judgment translates into ranking decisions.
 - Participants independently rank 100 AP news headlines, then merge rankings collaboratively, stating values and trade-offs.
 - These transcripts will be coded for value-operationalization.

Provider Fairness: Participatory Co-Design

Challenge: Algorithmic fairness metrics are designed by developers, not the providers they affect, resulting in mathematically defined "fairness" that fails in practice.

- Designed and facilitated 4 virtual focus groups with 13 content providers (social media creators and dating app users) to co-design fairness goals, definitions, and metrics from the provider perspective.
- Found that providers experience algorithmic unfairness in three distinct ways i.e., underexposure, overexposure, and lack of agency over platform curation, none of which are captured by standard fairness definitions.
- Showed that mathematical fairness does not guarantee real-world equity, and that conflicting stakeholder definitions create implementation failures when fairness is engineered without provider input. Published at FAccT 2024.

PAPER PUBLICATIONS/PRESENTATIONS

- **Satwani, A.** (2026). *Aligning Algorithms with Axiology: Operationalizing Journalistic Values in News Recommender Systems*. RecSys 2026 Doctoral Symposium (Accepted).
- **Satwani, A.,** & Burke, R. (2026). *Personalizing Diversity in News Recommendation: A User Study*. RecSys FRAME Workshop 2026 (Accepted).
- Higley, K., Burke, R., Ekstrand, M., Kluver, D., Knijnenburg, B., Malthouse, E., Hasan, M., Iqbal, M., Parajuli, S., **Satwani, A.,** Sristy, Z., Suna, R., & Konstan, J. (2026). *Designing HCAI Field Experiment Research Platforms: The POPROX News Experience*. International Journal of Human-Computer Interaction 2026 (Under Review).
- Smith, J. J., **Satwani, A.,** Burke, R., & Fiesler, C. (2024). *Recommend me? Designing fairness metrics with providers*. Proceedings of the 2024 ACM Conference on Fairness, Accountability, and Transparency.
- Alaqabawy, N., **Satwani, A.,** Volda, A., & Burke, R. (2023). *"It's just a robot that looks at numbers": Restoring journalistic voice in news recommendation*. INRA @ RecSys.
- Alaqabawy, N., **Satwani, A.,** & Burke, R. (2023). *The Journalistic Algorithm: A news recommender platform for public service journalism*. Broadcast Education Association.

INDUSTRY EXPERIENCE

Software Engineer III

Sept 2020 – Jun 2021

Walmart Global Tech, Bangalore, Karnataka, India

- Owned the full product lifecycle of AskME-HR chatbot I had originally built as an intern, scaling it to 10,000+ monthly active users; the product grew to support a dedicated sub-team.
- Added context-aware question recommendations and automated proactive notifications, personalizing the chatbot experience and improving self-service resolution without human escalation.
- Took the HR Merit Increment application (ReactJS + NodeJS) from requirements to production for 700 senior leaders, automating an annual process that previously required 3 months of manual Excel coordination between HRBPs and leadership.

- Built real-time dashboards giving leaders visibility into org-level headcount budgets and the ability to adjust individual increments directly, eliminating back-and-forth with HR and reducing planning cycle time.
- A/B tested feature additions continuously; all tested features passed and shipped to production.
- Designed a GraphQL API layer to enable a unified HR data lake across Walmart, giving multiple HR applications a single point of access to organizational data where previously each operated in isolation.

Software Engineer II

Jul 2018 – Sept 2020

Walmart Global Tech, Bangalore, Karnataka, India

- Built the HR Merit Web Application end-to-end: requirements, architecture, frontend (ReactJS), backend (NodeJS), infrastructure, and post-launch maintenance (as the sole engineer on the product).
- Scaled AskME's knowledge base and shipped NLP capabilities to production including sentiment analysis, speech-to-text, and multilingual support, converting what were initially POCs into live features used across the organization.
- Built PowerBI and Splunk dashboards to track usage analytics and feature adoption, creating the first visibility into how HR tools were being used at scale.

Software Engineering Intern

Jan 2018 – Jun 2018

Walmart Global Tech, Bangalore, Karnataka, India

- Built AskME, an HR self-service chatbot, from scratch as a solo intern project, handling backend integrations and deploying across Slack, Zoom, Teams, Webchat, and Chrome in collaboration with each platform's partner teams.
- The project was converted to a full enterprise product upon my return as a full-time engineer.

Software Engineering Intern

May 2022 – Aug 2022

Hitachi Energy, Remote, USA

- Designed and integrated a multi-horizon forecasting engine into the TRMTracker application using Neural Networks for short-term predictions and Multiple Linear Regression for medium and long-term energy parameter forecasting.
- Built REST APIs for real-time prediction delivery and SQL procedures for weighted training data generation; delivered as a functional prototype.

ACADEMIC PROJECTS

What's New: News Recommendation System | *Recommender Systems*

Aug 2023 – Dec 2023

- Implemented a news recommender using DKN, NRMS, and LSTUR on the MIND dataset, incorporating Maximal Marginal Relevance (MMR) for diversity. Evaluated recommendations using pairwise similarity-based diversity score, AUC, and NDCG.

Itinerary Optimizer | *Linear Programming*

Jan 2022 – May 2022

- Developed a cross-country itinerary optimizer using a Genetic Algorithm to plan routes covering points of interest, refueling stops, and restaurants. Handled constraints such as daily travel limits to generate feasible and efficient itineraries.

Product Feature Analysis Based on Customer Reviews | *Big Data Analysis*

Jan 2022 – May 2022

- Analyzed 1.2M customer reviews over 5 years using LDA topic modeling to extract product features. Identified 30 salient terms and distilled them into the 14 most relevant features per category, performing feature-specific sentiment analysis.

Patronizing and Condescending Language Detection | *Natural Language Processing*

Aug 2021 – Dec 2021

- Implemented a binary classification model using RoBERTa from HuggingFace to detect Patronizing and Condescending Language (PCL) for SemEval 2022 Task 4. Evaluated model performance using accuracy, recall,

precision, and F1 score. Leveraged transformers for text embeddings, demonstrating skills in NLP modeling and representation learning.

Movie Recommendation System | *Machine Learning*

Jan 2017 – May 2017

- Created a user-user collaborative filtering recommendation system using the MovieLens dataset. Computed similarity metrics with Euclidean distance and applied Laplace smoothing, evaluating performance using RMSE and NDCG.

FELLOWSHIPS, HONORS, AND AWARDS

- Lloyd Botway Fellowship (2023), University of Colorado, Boulder
- Outstanding TA Recognition Award (2023), University of Colorado, Boulder
- Outstanding Service Award (2023, 2024), University of Colorado, Boulder
- Bravo Award (2020), Walmart Global Tech

ACADEMIA EXPERIENCE

- Teaching Assistant – CSCI 4308 Software Engineering Project Management/Mentoring, Department of Computer Science, University of Colorado, Boulder
- Teaching Assistant – CSCI 3308 Software Development Methods and Tools, Department of Computer Science, University of Colorado, Boulder
- Ph.D. Representative – CS Graduate Professional Development Club, 2023, Department of Computer Science, University of Colorado, Boulder
- Vice Chair – CS Graduate Student Association, 2022, Department of Computer Science, University of Colorado, Boulder

SKILLS

Programming & ML: Python, Java, C++, R, Scala, NodeJS, ReactJS, JavaScript, HTML, Apache Spark, Hadoop, Kafka, SQL, NoSQL, PyTorch, TensorFlow, Scikit-learn, Pandas, NumPy, Surprise, Hugging Face Transformers

Recommender Systems: Collaborative, content-based, hybrid, ranking, sequence, graph-based, value-based, fairness-aware, LLM-based; Metrics: Accuracy(NDCG, MRR, RMSE), Non-Accuracy(diversity, novelty, coverage, calibration)

Cloud & Infrastructure: AWS, GCP, Azure, Docker, Kubernetes, Jenkins, OneOps

REFERENCES

Dr. Robin Burke

Doctoral Advisor, University of Colorado Boulder

Professor of Information Science and Computer Science

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